

Type: DSC 2-3 (Minor) B.Sc(ECS)-II (Semester III) Course Title: Statistics for Data Science-I (Paper Code:)		
Credits: Theory – (2)		Practical's – (1)
Total Lectures: 30 Hrs.		Contact Hrs. (L) : 2
University Evaluation: 30 Marks		Internal Evaluation: 20 Marks
Course Outcomes:		
1.	Students will demonstrate proficiency in statistical analysis of data.	
2.	Students will develop the ability to build and assess data-based models.	
3.	Students will execute statistical analyses with professional statistical software.	
4.	Students will demonstrate skills in data management.	
5.	Students will apply data science concepts and methods to solve problems in real-world contexts and will communicate these solutions effectively	
Unit	Content	Lect.
I	Population and sample: Concept of the statistical population with illustration, Concept of the statistical sample with illustration. Sampling methods: Simple Random Sample(SRS), Simple Random Sample With Replacement (SRSWR), Simple Random Sample Without Replacement (SRSWOR), stratified and systematic random sampling. Data condensation and graphical methods: Raw data- attribute, variable- discrete and continuous, constant. Classification- definition, objectives, principles of classification. Construction of frequency distribution, cumulative frequency distribution. Graphical representation: histogram, ogives, pie chart. Measures of Central Tendency: Concept of central tendency, the criterion for good measures of central tendency. Arithmetic Mean(A.M.): definition computation for ungrouped and grouped data, combined mean, weighted mean, effect of change of origin and scale, numerical examples, merits and demerits. Median: definition, computation for ungrouped and grouped data, graphical determination, numerical examples, merits and demerits. Mode: definition, computation for ungrouped and grouped data, graphical determination, numerical examples, merits and demerits.	15

	Quartiles: definition, computation for ungrouped and grouped data, graphical determination, Numerical problems. Measures of Dispersion: Concept of dispersion, absolute and relative measures, Range: definition, computation for ungrouped and grouped data, Coefficient of range, numerical examples, merits, and demerits. Standard Deviation(S.D.) and Variance: definition, computation for ungrouped data and grouped data. Coefficient of Variation(C.V.): definition, computation for ungrouped data and grouped data, combined S.D. effect of change of origin and scale, merits and demerits, Numerical problems.	
II	Correlation and Regression: Bivariate data, Concept of correlation, types of correlation, cause and effect relation. Methods of Studying Correlation: a) Scatter Diagram Method: definition, interpretation. b) Karl Pearson's coefficient of correlation (r) definition, limits for 'r', interpretation of r, effect of change of origin and scale, numerical examples. Regression: concept, definition, lines of regression, Derivation of lines of regression by least square method, properties of regression coefficients, interpretation of regression coefficients. Non-linear regression: fitting of lines of regression, fitting of second degree curve ($Y = A + BX + CX^2$), fitting of exponential curve ($Y = AB^x$), numerical problems.	15
Reference Books:		
1.	Fundamentals of Statistics by Goon Gupta, Das Gupta.	
2.	Statistical Methods by S. P. Gupta	
3.	Business Statistics by S. Shaha	
4.	Fundamentals of Mathematical Statistics by Kapoor and Gupta	
5.	Programmed Statistics by B. L. Agarwal	